

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Langflow, IBM Orchestrate and IBM Models”

Project Tile – [BlogPulse – Agentic AI Writing Assistant for Bloggers]

Domain of Project – Content Creation / Media & Publishing (Blogging Assistant)

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Problem Statement -

Problem Statement : Agentic AI Writing Assistant for Bloggers

The Challenge

Bloggers today face an overwhelming flood of scattered information—news updates, research articles, social media trends, and audience feedback. This content is fragmented, quickly outdated, and difficult to **organize, making it hard for writers to focus on creating impactful blogs**. The challenge is not just to collect ideas but to **synthesize, prioritize, and guide writers toward high-quality content creation**.

The Objective

Build an **agentic AI application** that acts as an **intelligent writing assistant** for bloggers. The solution should:

- **Fuse Disparate Sources** – Collect real-time data from multiple channels (news sites, research papers, social media, comments) and synthesize into clear summaries with actionable blog topic suggestions.
- **Enable Multimodal Content Curation** – Accept user inputs (text, images, or voice notes), refine ideas, and recommend blog structure, tone, or supporting visuals.
- **Create a Predictive & Agentic Layer** – Analyze engagement patterns to predict trending topics and suggest optimal publishing time, hashtags, or angles for maximum reach.
- **Visualize the Writing Pulse** – Provide a clean, real-time dashboard showing curated topics, sentiment analysis, and engagement forecasts to help writers align with audience interests.
- **Multi-Agent System** –
 - **Source Fusion Agent** (collects and synthesizes news, research, and social data)
 - **Content Curation Agent** (refines multimodal inputs into blog structure and tone)
 - **Trend Prediction Agent** (forecasts trending topics and optimal reach)
 - **Engagement Analytics Agent** (sentiment analysis and engagement tracking)
- **Writing Pulse Dashboard** – Real-time view of curated topics, sentiment analysis, and engagement forecasts.

Proposed Solution -

Proposed Solution: BlogPulse – Agentic AI Writing Assistant

The proposed solution is a **multi-agent agentic AI writing assistant** built using **IBM Langflow / IBM Orchestrate and IBM Granite Models** to fuse scattered content sources into actionable blog insights. The system addresses the challenges of fragmented information, content curation, and audience engagement through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch real-time content from trusted sources such as news sites, research papers, and social media. This data is embedded and stored in a vector database, enabling the system to retrieve and synthesize clear summaries and blog topic suggestions using Granite models.

The system consists of four key agents:

Source Fusion Agent retrieves and synthesizes news, research, and social content into clear summaries using RAG.

Content Curation Agent accepts multimodal user inputs (text, images, voice notes), refines ideas, and recommends blog structure, tone, and supporting visuals.

Trend Prediction Agent analyzes engagement patterns to predict trending topics and suggests optimal publishing time, hashtags, and angles for maximum reach.

Engagement Analytics Agent tracks sentiment and engagement data, feeding real-time insights back into the dashboard for writers.

The solution includes a **clean, real-time “Writing Pulse” dashboard** (built with Streamlit or React) that visualizes curated topics, sentiment analysis, and engagement forecasts to help writers align with audience interests.

IBM Langflow / IBM Orchestrate is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent writing assistant** that helps bloggers cut through information overload with synthesized insights, predictive trend guidance, and real-time engagement analytics.

Technology Used

- **IBM Langflow / IBM Orchestrate** – For designing and orchestrating the multi-agent writing-assistant workflows visually
- **IBM Granite model** – for eg – ibm-granite-3-2-8b – For natural language understanding, summarization, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the writing assistant efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the agentic AI application.
- **RAG**– RAG to **fetch real-time news, research, and social content first, then generate synthesized summaries and blog suggestions**, making the writing assistant more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based retrieval of news, research, and social content

Required Files created using IBM Langflow / IBM Orchestrate -

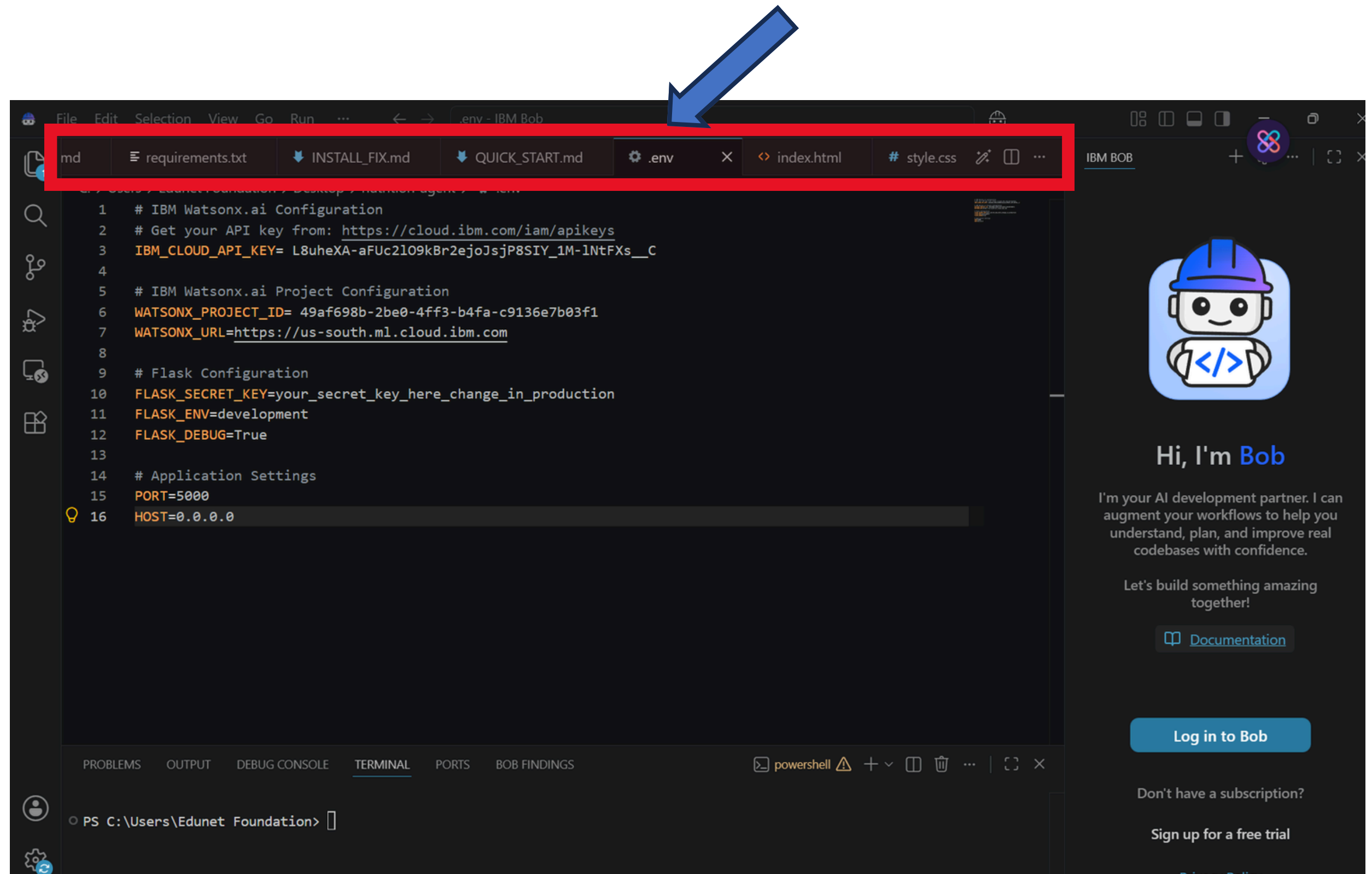
1. App.py

2. Requirements.txt

3. Style.css

4. .env

5. Index.html



IBM Langflow / Orchestrate Screenshots:

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File

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Selection

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.env - IBM Bob

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C: > Users > Edunet Foundation > Desktop > nutrition-agent > .env

IBM Watsonx.ai Configuration

Get your API key from: https://cloud.ibm.com/iam/apikeys

IBM_CLOUD_API_KEY= L8uheXA-aFUc2l09kBr2ejoJsjP8SIY_1M-1NtFXs__C

IBM Watsonx.ai Project Configuration

WATSONX_PROJECT_ID= 49af698b-2be0-4ff3-b4fa-c9136e7b03f1

WATSONX_URL=https://us-south.ml.cloud.ibm.com

Flask Configuration

FLASK_SECRET_KEY=your_secret_key_here_change_in_production

FLASK_ENV=development

FLASK_DEBUG=True

Application Settings

PORT=5000

HOST=0.0.0.0

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PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

PS C:\Users\Edunet Foundation>

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Screen Reader Optimized

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Spaces: 4

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Bob - Settings

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IBM BOB



Hi, I'm Bob

I'm your AI development partner. I can augment your workflows to help you understand, plan, and improve real codebases with confidence.

Let's build something amazing together!

Documentation

Log in to Bob

Don't have a subscription?

Sign in failed: Cancelled by user.

File Edit Selection View Go Run Terminal Help

← → bob test

Upgrade your plan

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EXPLORER

BOB TEST

.venv

static

script.js

style.css

templates

index.html

.env.example

.gitignore

app.py 8

DEPLOYMENT.md

Procfile

README.md

requirements.txt

runtime.txt

SETUP_GUIDE.md

requirements.txt

.env.example

app.py 8 X

index.html

script.js

README

app.py > get_watsonx_model

73 # =====

74 # IBM Watsonx.ai Configuration

75 # =====

76

77 def get_watsonx_model() -> Model | None:

78 """Initialize and return IBM Watsonx.ai model"""

79 try:

80 credentials: Credentials = Credentials(

81 url=os.getenv('IBM_WATSONX_URL', 'https://au-syd.ml.cloud.ibm.com'),

82 api_key=os.getenv('IBM_CLOUD_API_KEY'),

83)

84

85 project_id: str | None = os.getenv('IBM_WATSONX_PROJECT_ID')

86

87 model: Model = Model(

88 model_id='ibm/granite-guardian-3-8b',

89 params={

90 GenParams.DECODING_METHOD: "greedy",

91 GenParams.MAX_NEW_TOKENS: 900,

92 GenParams.MIN_NEW_TOKENS: 50,

93 GenParams.TEMPERATURE: 0.7,

94 GenParams.TOP_K: 50,

95 GenParams.TOP_P: 1

96 },

97 credentials=credentials,

98 project_id=project_id

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

BOB FINDINGS

requests==2.31.0

- urllib3==2.7.0

+ urllib3==2.1.0

+ werkzeug==3.0.1

+ zipp==4.1.0

warning: `ibm-watson-machine-learning==1.0.335` is yanked (reason: "Causes errors on py 3.9")

(.venv) PS D:\OneDrive - Edunet Foundation\Desktop\bob test>

powerShell

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CODEX

IBM BOB

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Create an AI-powered Nutrition ... 42k / 200k 1.17 ✕

All tasks completed! 13/13

1. Install dependencies: `pip install -r requirements.txt`

2. Copy `.env.example` to `.env` and add IBM credentials

3. Run: `python app.py`

4. Open: `http://localhost:5000`

Customization:

The `AGENT_INSTRUCTIONS` section in `app.py` allows you to easily customize:

- Agent tone and personality
- Diet specializations
- Indian food preferences
- Safety rules and limitations
- Response formatting

All files are production-ready with comprehensive error handling, security best practices, and deployment configurations for Heroku, IBM Cloud, AWS, and Docker!

Start New Task

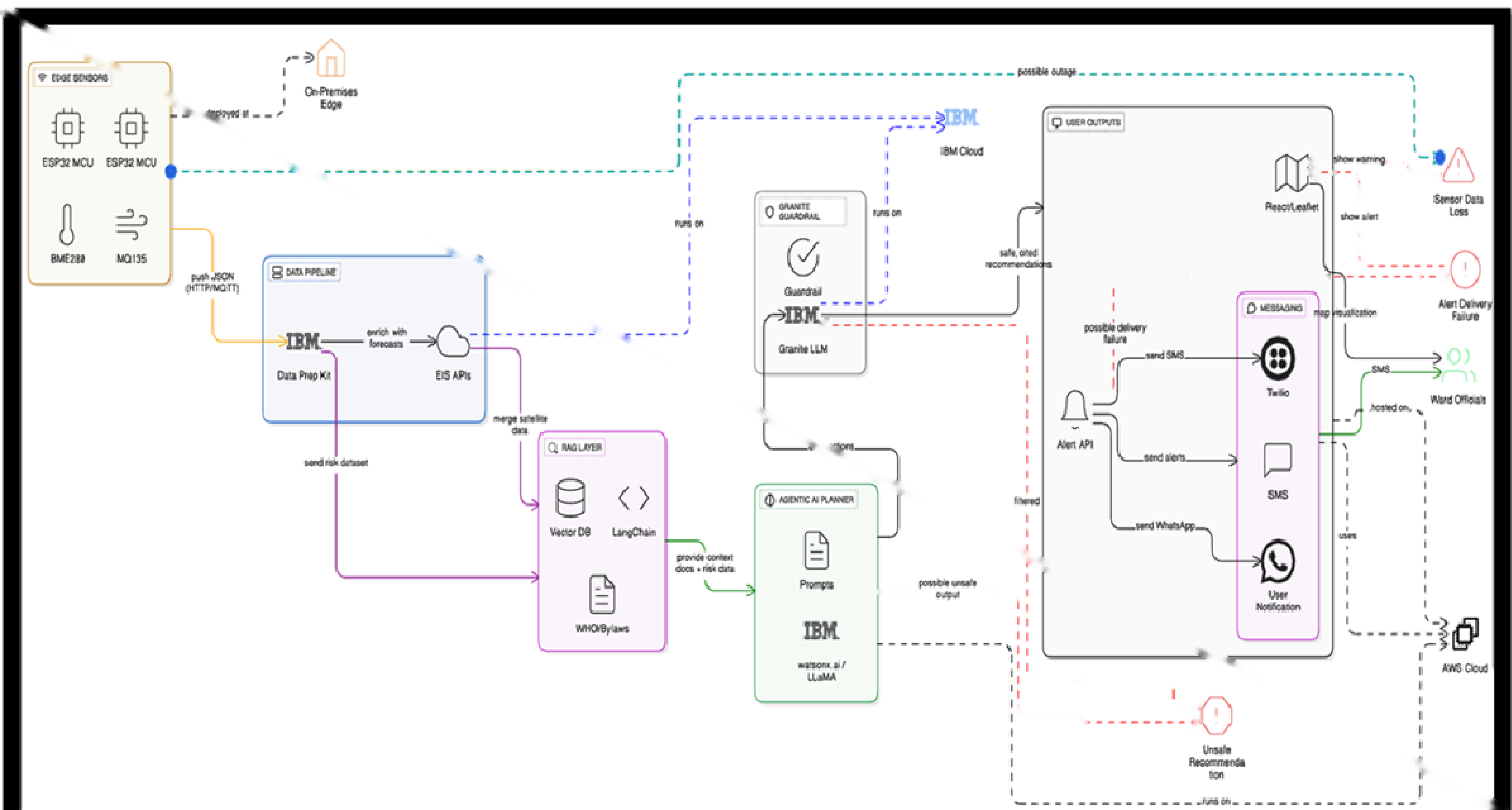
What's on your mind?

(@ to add context, / for commands, hold shift to drag in files/images)

Code ⌵

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Architecture blueprint for BlogPulse:



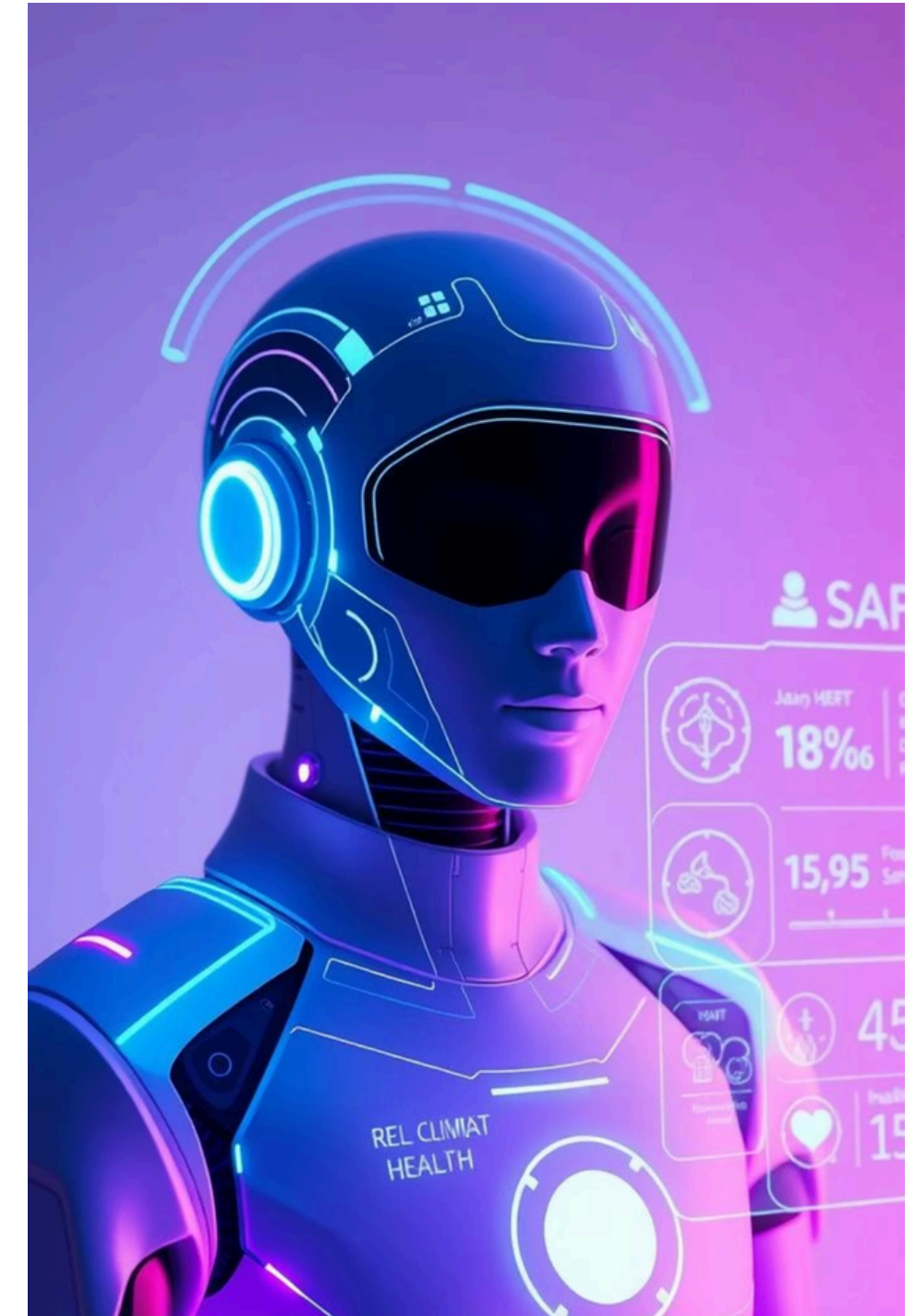
Role of Agentic AI in the solution

Role of Agentic AI in BlogPulse

Agentic AI enables BlogPulse to function as an intelligent, autonomous writing assistant rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver synthesized, trend-aware content guidance.

Each agent performs a specific role—such as fusing scattered sources, curating multimodal content, predicting trends, and analyzing engagement—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate, trustworthy results. Agentic AI also brings **context awareness**, understanding a writer's tone, audience, and past content to generate tailored blog suggestions. It supports **real-time adaptation**, continuously updating topic and engagement predictions based on live data streams. Additionally, agents communicate with each other, enabling better decision-making and higher-quality content outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital writing partner** that helps bloggers consistently create high-impact content.



Token Usage

[Add your token usage count here]

IBM **Bob**

Organization

bob-001 (us-east)

Subscription

API Keys

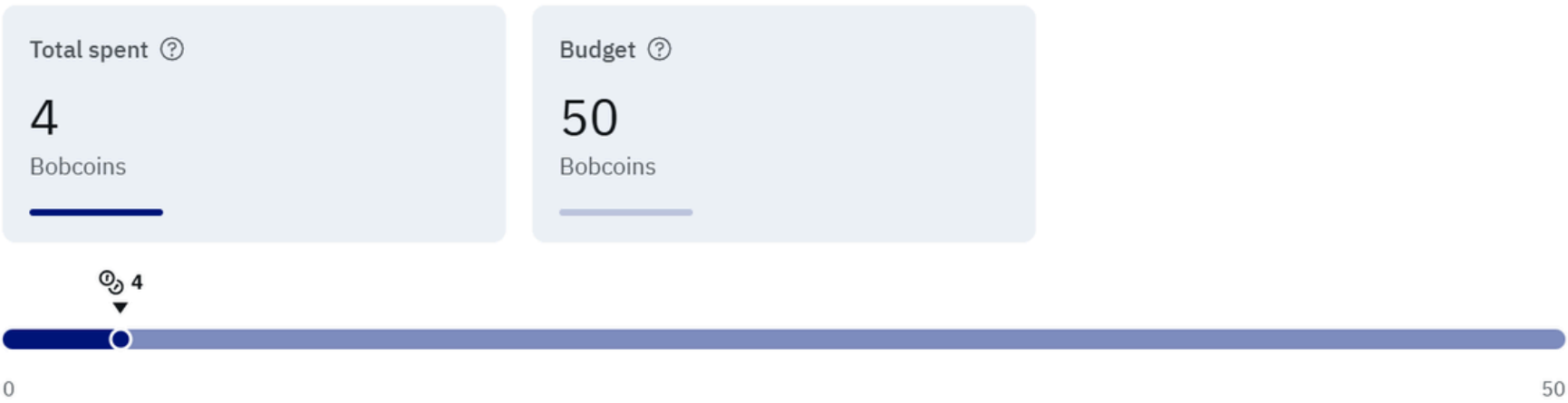
J

User
jeemail9996@gmai...

Account	jeemail9996@gmail.com	
Plan	Trial	Manage my subscription
Region	United States (East)	

Monthly Bobcoin usage

Your budget resets every month and renews on 9 October 2026

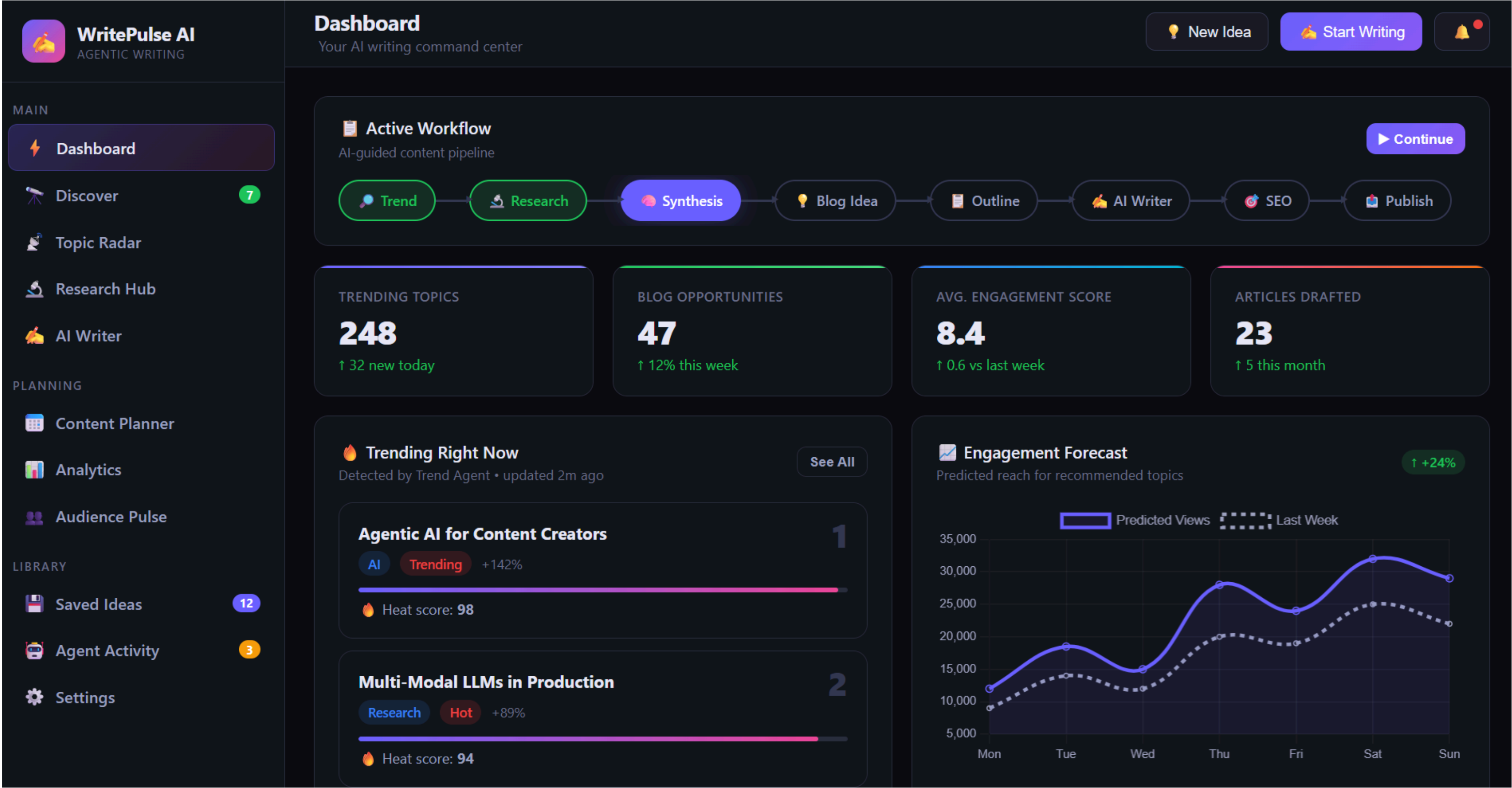


Your trial expires on 9 October 2026

Project Output Screenshot

[illegible]

Project Output Screenshot



Project Output Screenshot



WritePulse AI

AGENTIC WRITING

MAIN

 Dashboard

 Discover

7

 Topic Radar

 Research Hub

 AI Writer

PLANNING

 Content Planner

 Analytics

 Audience Pulse

LIBRARY

 Saved Ideas

12

 Agent Activity

3

 Settings

Discover

Find emerging trends across all sources

New Idea

Start Writing



Discover via:

Text

Image

Voice

Enter a topic, paste a URL, describe an idea, or upload an image / record voice note...

Discover

All Sources

News

Research

Reddit

Social

Blogs

 Emerging Signals

Detected by Research Agent across 12 data sources

"Auto-blogging with agents" searches up 340%

Google Trends 8m ago

98

→ Explore

r/MachineLearning viral post: "I replaced my whole research process with 3 AI agents"

Reddit 23m ago

94

→ Explore

New ArXiv paper: "Autonomous content generation benchmarks 2025"

Research 1h ago

87

→ Explore

@sama tweets about agentic frameworks — 84K engagements

82

 Source Cards

Top-quality content sources being monitored

TechCrunch

News • 24 items

Hot

ArXiv

Research • 47 items

Rising

Reddit

Community • 89 items

Very Active

Twitter/X

Social • 340 items

Trending

Substack

Blog • 16 items

Steady

Novelty and Uniqueness

BlogPulse stands out by combining **Agentic AI, RAG, and predictive personalization** into a single intelligent writing companion.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate, context-aware writing guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy topic suggestions.

Hyper-Personalization – Tailors blog structure, tone, and visuals based on a writer's style, audience, and goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on live engagement data and audience sentiment.

Multi-Modal Input – Supports text, voice notes, and images for richer, more flexible content curation.

Predictive Trend Focus – Goes beyond content curation to forecast trending topics and optimal publishing angles.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital writing partner**, not just a static topic-suggestion tool.

Git Hub Link

Github link - <https://github.com/dubble911/ibm-agentic-ai->

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Future Scope

1. Integration with Publishing Platforms & CMS –

BlogPulse can be integrated with platforms such as WordPress, Medium, and Ghost to auto-schedule posts and pull live performance metrics. This will enable the agent to provide more accurate and dynamic topic recommendations based on actual publishing outcomes.

2.AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility for bloggers. Writers can dictate ideas and interact naturally in their preferred language, making the solution more inclusive for diverse, global audiences.

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Certificates Earned:

IBM SkillsBuild – Getting Started with Artificial Intelligence

In recognition of the commitment to achieve professional excellence



rishikant dubey

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Issued on: Sep 09, 2026

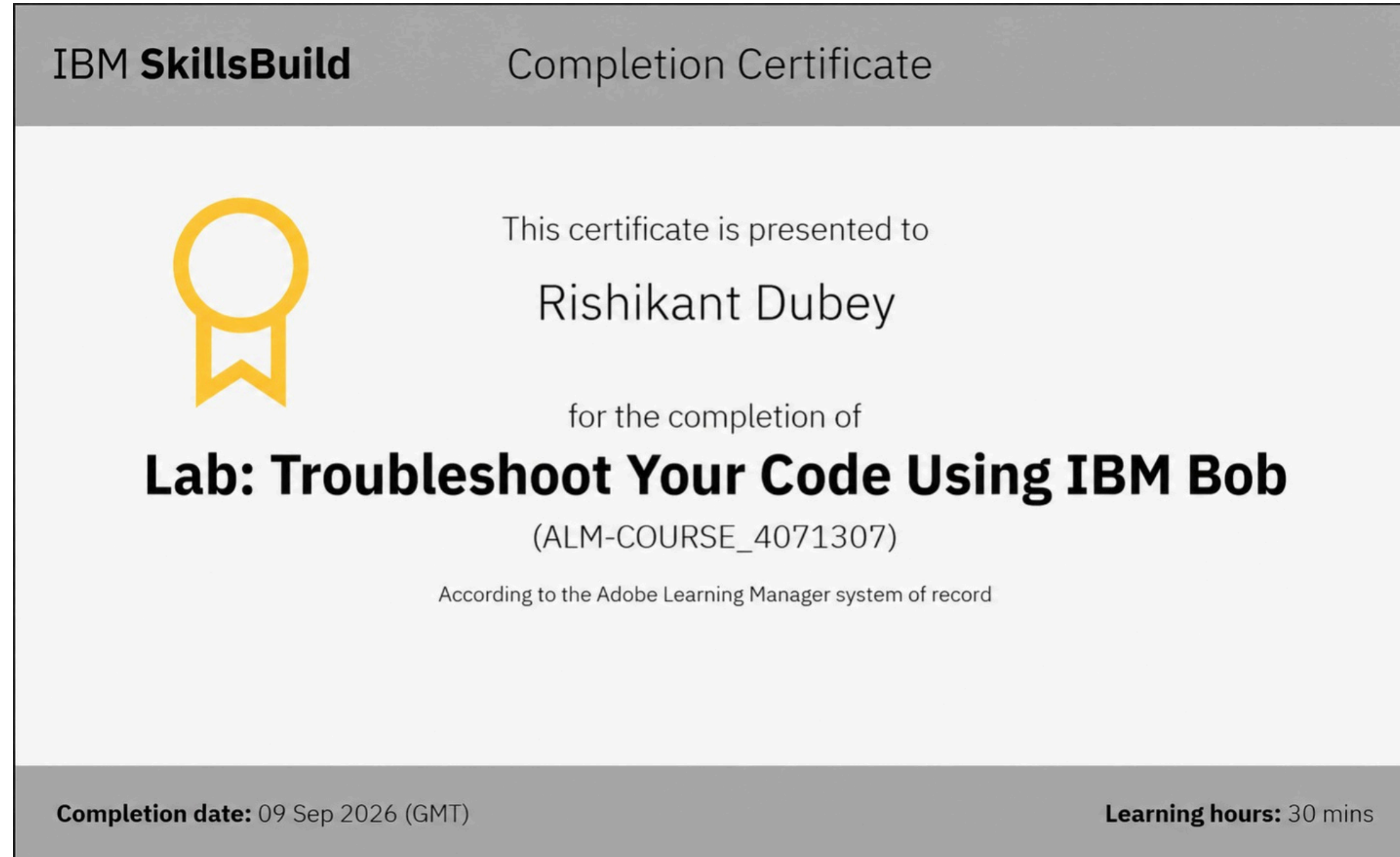
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/152ae6da-bfdb-43d6-a6cd-eb4ade54397d>



Certificates Earned:

IBM SkillsBuild – Lab: Troubleshoot Your Code Using IBM Bob



Thank You!

Thank you for your time and interest.